ACCELERATING THE FERMI-LOWDIN ORBITAL DESCRIPTOR OPTIMIZATION FOR SELF-INTERACTION FREE DFT CALCULATIONS

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DEDICATION

To my father, who is the inspiration behind my every progress.

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Thanks to my colleagues at the FLO-SIC group for their support and help. And last, but not the least, I want to thank the faculty members and staffs of CMU physics department for their support whenever it was needed.

ABSTRACT

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by Md Nageeb Bin Zaman

The Fermi-Löwdin Self Interaction Correction (FLO-SIC) method was introduced to address the shortcomings of standard density functional approximation calculations by removing the spurious electron self-interaction. Within this method, a set of Fermi orbital descriptors (FODs) need to be optimized to minimize the energy. Current FOD optimization algorithms require large numbers of steps and the number of steps increases with systems size, or equivalently, as the number of FODs increases. Moreover, every step in this optimization process requires the time-consuming evaluation of the single-particle energies and potential matrix elements. In this thesis, I analyze a strategy where I fix the potential during the FOD optimization process, making the calculation of potentials at every step of the optimization unnecessary. Bypassing the need for calculating potentials at each step would make the optimization process more efficient and less time consuming. I will present proof-of-concept results for small molecules and compare with existing schemes.

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CHAPTER 1

INTRODUCTION

One of the central problems in theoretical physics and chemistry is to solve the depiction of the structure and properties of many-electron system. According to non-relativistic quantum theory, the state of any given system is represented by a wave function Ψ and it obeys the Schrödinger equation,

$$\hat{H}\Psi = E\Psi\,. \tag{1.1}$$

For a many electron system subject to an electrostatic interaction with nuclei, this equation becomes,

$$\left[\hat{T} + \hat{V} + \hat{U}\right]\Psi = E\Psi \tag{1.2}$$

or,

$$\left[\frac{-\hbar^2}{2m}\sum_{i=1}^{\mathbb{N}}\nabla_i^2 + \sum_{i=1}^{\mathbb{N}}V(r_i) + \sum_{i<j}U(r_i, r_j)\right]\Psi = E\Psi\,, \tag{1.3}$$

here N is the number of electrons, $\hat{T}$ is the kinetic energy term, $\hat{V}$ is the Coulomb interaction among electrons and nuclei, and $\hat{U}$ represents the Coulomb repulsion between electrons.

The solution to the Schrödinger equation is a fully anti-symmetric wave function of the form $\phi = \phi(r_1, r_2, ..., r_N)$. So the number of spatial variables to be calculated to obtain the correct solution is 3N, and as the number of variables increases, the complexity of the problem grows exponentially. This scaling makes the computational problem very expensive for systems of increasing size.

1.1 Density Functional Theory (DFT)

Density Functional Theory is the most popular also most successful electronic structure method in modern solid state physics and material science. The entire field of DFT is based on two fun-

damental mathematical theorems given by Hohenberg and Kohn [1] and also the derivation of a group of equations by Kohn and Sham[2].

First Hohenberg-Kohn Theorem: The ground state energy from Schrödinger's equation is a unique functional of the electron density. The theorem states that, between ground-state wave function and ground-state electron density there exists a one-to-one mapping, so now we can express ground state energy E, as $E[n(r)]$, where $n(r)$ is the electron density,

$$n(\vec{r}) = N \int d^3r_2 \int d^3r_3 .... \int d^3r_N \Psi^*(\vec{r}, \vec{r}_2, ..., \vec{r}_N)\Psi(\vec{r}, \vec{r}_2, ..., \vec{r}_N) \, . \tag{1.4}$$

This opens the possibility of finding solution of the Schrödinger by using electron density which is a function of three spatial variables, rather than using the wave function which is a function of 3N spatial variables.

Second Hohenberg-Kohn Theorem: The electron density that minimizes the energy of the overall functional is the true electron density corresponding to the full solution of the Schrödinger equation. This means that if we knew the actual functional form, we could minimize the energy from the functional by varying the electron density. The energy functional can be expressed as

$$E[n] = E_{known}[n] + E_{XC}[n] \, . \tag{1.5}$$

Here, $n = \sum_i \phi_i^* \phi_i$

and,

$$E_{known}[\Psi_i] = \frac{-\hbar^2}{2m} \sum_i \int \phi_i^* \nabla^2 \phi_i \, d^3r + \int V_{Ne}(\vec{r}) n(\vec{r}) \, d^3r + \frac{e^2}{2} \iint \frac{n(\vec{r}) n(\vec{r}')}{|\vec{r} - \vec{r}'|} \, d^3r \, d^3r' + E_{ion} \, . \tag{1.6}$$

The terms on the right-hard side of Eq. 1.6 are the kinetic energy of the e^-, the Coulomb interaction between e^- and nuclei, Coulomb interaction between pairs of e^-, and the Coulomb nuclear pairwise interaction, respectively.

In the energy functional of Eq. 1.5, E_{XC} is the exchange-correlation functional. The exchange-correlation functional is defined is such a way that it includes all the quantum mechanical effects that exclided from E_{known}.

Since we are looking for the ground-state electron density, according to Kohn and Sham [2], the many-body Schrödinger equation can be reduced to one-electron equations,

$$\left[\frac{h^2}{2m}\nabla^2 + V_{Ne}(\vec{r}) + V_H(\vec{r}) + V_{XC}(\vec{r})\right]\Psi_i(\vec{r}) = \varepsilon_i\Psi_i(\vec{r})\,. \tag{1.7}$$

The solutions of the Kohn-Sham equations are single-electron wave functions. And these solutions depend on only three variables. Here, V_H is the Hartree potential which describes the Coulomb repulsion between one electron and the total electron density , which includes the non-physical self-interaction of the electron, and V_{XC} defines the exchange and correlation contributions to the Kohn-Sham equations.

We can find the ground state by minimizing the energy by finding a self-consistent solution to a set of single particle equations. To solve the Kohn-Sham equation to get the exact density, we must have the exact form of exchange-correlation functional $E_{XC}[n]$. However, the exact form of the functional is not known. Over the course of decades several approximate functionals have been developed, and each of them yields reasonably accurate results for certain cases[3–5].

The first and the simplest exchange-correlation functional is the local density approximation (LDA). LDA uses only the local density as an ingredient. After the LDA the best known class of density functionals is the generalized gradient approximation (GGA). In case of GGA the exchange-correlation energy density is considered to be a functional of not only the local spin densities, but of the local spin density gradients as well[6]. There are a lot of distinct GGA functionals as different functionals employ different parameterizations of the energy in terms of the density and its gradient. Among them two most used functionals used for calculations in-

volving solids are the Perdew-Wang functional (PW91)[7, 8] and Perdew-Burke-Ernzerhof functional(PBE)[6].

If the exact exchange-correlation functional were known, self-interaction could have been eliminated exactly. In equation 1.6 the Hartree energy term incorporates the interaction of each electron with itself. The exact functional would cancel this self-interaction. But since in practice approximate exchange-correlation functionals are used, the self-interaction cannot be eliminated exactly resulting in the presence of self-interaction error energy E_{SIE} in the functional. As a result these approximate functionals are not exact for one-electron density. There are also some problems associated with SIE, like too high orbital energies, too low chemical reaction barriers etc. Eliminating E_{SIE} is the aim of self-interaction correction (SIC). The self-interaction correction method most frequently applied today is one proposed by Perdew and Zunger.

1.2 Perdew-Zunger Self-Interaction Correction

To remove electron self-interaction from DFT in 1981, Perdew-Zunger self-interaction correction (PZ-SIC)[9–11] was introduced. In PZ-SIC the spurious self-interaction energy is subtracted out from the total energy functional on an orbital by orbital basis,

$$E^{DFT-SIC} = E^{DFT}[n^{\alpha}, n^{\beta}] - \sum_{i\sigma}(E_{XC}[n_i^{\sigma}, 0] + E_H[n_i^{\sigma}])\,, \tag{1.8}$$

where, E^{DFT} is the DFT energy functional, E_{XC} is the exchange-correlation functional, E_H is the Hartree or Coulomb energy and n_i^{σ} are single orbital densities of spin σ,. Analogous to the Kohn-Sham equations, Schrödinger-like DFT-SIC equations are derived using a standard variational method, yielding

$$(\hat{H}_{\sigma}^{DFT} + V_{i\sigma}^{SIC})\phi_{i\sigma} = \hat{H}_{i\sigma}^{DFT-SIC}\phi_{i\sigma} = \sum_{j=1}^{M_{\sigma}} \lambda_{ji}^{\sigma}\phi_{j\sigma}\,. \tag{1.9}$$

Here $\hat{H}_{\sigma}^{DFT}$ is the DFT Hamiltonian, $V_{i\sigma}^{SIC}$ is the SIC potential of the orbital i, and $\hat{H}_{i\sigma}^{DFT-SIC}$ is the corresponding SIC Hamiltonian. To ensure orthogonality between the M_{σ} occupied orbitals, Lagrange multipliers λ_{ji}^{σ} are introduced in the variational procedure.

In case of PF-SIC, thr total energy $E^{DFT-SIC}$ and the DFT-SIC equations are explicitly orbital dependent. This is a challenging feature because, this means that each orbital densities must be optimized to get the lowest $E^{DFT-SIC}$. As a result, the method has suffered from the computational demands tied to it[12–15]. Furthermore, this procedure may not be entirely size-extensive. In this case size-extensivity means that adding the total energies of spatially well-separated electron densities yields the total energy of a system containing these densities.

1.3 Fermi-Löwdin Self-Interaction Correction (FLOSIC)

The Fermi-Löwdin Self-Interaction Correction, or FLO-SIC, [12, 14, 16] is a recent implementation of PZ-SIC that offers potential computational advantages. The key idea behind FLO-SIC is to implement localized Fermi orbitals (FO)[17, 18] in the PZ energy functional. The FLO-SIC method reintroduces unitary invariance and size-extensivity in self-interaction free DFT and also accomplishes numerical scaling equal to that of the unmodified Kohn-Sham approach. Compared to PZ-SIC, FLO-SIC does not evaluate any possible unitary transformation of the occupied Kohn-Sham orbitals in order to obtain the total energy minimum. Therefore, solving the computationally demanding $O(N^2)$ localization equations [19] is not required, and instead the localized orbitals are constructed in a fixed way. For a system of N electrons, first we need to transform the occupied Kohn-Sham orbitals into Fermi orbitals(FO). The transformation is defined as

$$F_{i\sigma}(r) = \sum_{j=1}^{M_{\sigma}} \frac{\psi_{j\sigma}^{*}(\mathbf{a}_{i\sigma})\psi_{j\sigma}(\mathbf{r})}{\sqrt{\sum_{j=1}^{M_{\sigma}} |\psi_{j\sigma}(\mathbf{a}_{i\sigma})|^2}}, \tag{1.10}$$

where $\psi_{j\sigma}$ is any set of orbitals that spans the occupied orbital space (*i.e.* $\sum_{j=1}^{M_\sigma} |\psi_{j\sigma}(\mathbf{a}_{i\sigma})|^2 = n_\sigma(\mathbf{r})$). The $a_{i\sigma}$ are called the Fermi-orbital descriptors (FODs). The FODs are M_σ distinct points in space that determine the transformation into Fermi orbitals. The FODs can be seen as quasi-classical electron positions and they usually can be approximately guessed based on physical and symmetry arguments and they apparently contribute to the generation of the Fermi-orbitals as a parameter[12, 16, 20]. Hence, for transforming M^σ Kohn-Sham orbitals into Fermi orbitals only M^σ of these parameters are needed compared to $(M^\sigma)^2$ entries of the transformation matrix [19] in the PZ-SIC formalism. This gives us significant reduction in computational effort compared to PZ-SIC. The norm of each FO is one, but they are not orthogonal. To guarantee orthonormal localized orbitals, the Löwdin method of symmetric orthonormalization[21] is applied to the FOs to obtain a set of orthonormal Fermi-Löwdin orbitals (FLOs), which are then used in the FLO-SIC method.

In the FLO-SIC methodology, we need to optimize both the total density and the FLOs to minimize the total energy $E^{DFT-SIC}$. The FLOs are optimized by rearranging the FOD positions. FOD positions can be optimized by utilizing the gradients of the total energy with respect to the a_i which are also referred to as the FOD forces (although these gradients are not associated to a physical force). FOD forces are given by [22]

$$\frac{dE^{SIC}}{da_i} = \sum_{kl} \langle\psi_l|V_k^{SIC}|\psi_k\rangle \left[\left\langle \frac{d\psi_k}{da_i}\middle|\psi_l\right\rangle - \left\langle \frac{d\psi_l}{da_i}\middle|\psi_k\right\rangle\right] . \tag{1.11}$$

Here, V_k^{SIC} is the SIC potential, ψ are the Fermi-Löwdin orbitals and a_i are the FODs.

At the minimum of $E^{DFT-SIC}$, the local orbitals satisfy equation 1.9 consistently and the forces on the FODs are zero. A brief outline of the self-consistency procedure of FLO-SIC calculations is presented below.

First, to generate the DFT Hamiltonian and SIC potentials and to compute the the matrix elements $\langle\psi_j|H_i|\psi_i\rangle$ the current guess of the local orbitals is used. While the index i extends only over the

occupied orbitals, the index j extends over both occupied and unoccupied orbitals.

Using a set of sequential 2×2 unitary rotations with the unoccupied orbitals the occupied orbitals are updated in order to remove the projection of $H_i |\psi_i\rangle$ onto the unoccupied orbitals. Then an updated set of local orbitals is created using the updated occupied orbitals used with the FODs. In the next iteration an updated DFT-SIC energy is evaluated and the DFT Hamiltonian and SIC potentials are calculated.This process is repeated (using the same set of FODs) until the total energy converges and self-consistency is reached. Symmetry considerations were not employed in these calculations. After the density is converged, the FOD are relaxed at constant density using a gradient-based optimizer that returns an set of FOD positions which are updated. These steps are repeated until the maximum FOD force reaches a specified tolerance.Depending on the system, it typically requires tens to hundreds of steps to reach the tolerance threshold, using this strategy. The convergence behavior is strongly dependent on the initial FOD positions and also the particular system. In Fig.1.1 a flowchart is given which summarizes the energy minimization in FLO-SIC.

In practice, this algorithm may require hundreds of steps to optimize the FOD positions, and the number of steps required will grow as the number of FODs increases. Thus, if we have a large system with a large number of FODs the FLO-SIC code using this algorithm will take a lot of time to optimize the FOD positions. It is quite evident that it is important to make the FOD optimization process as efficient as possible.

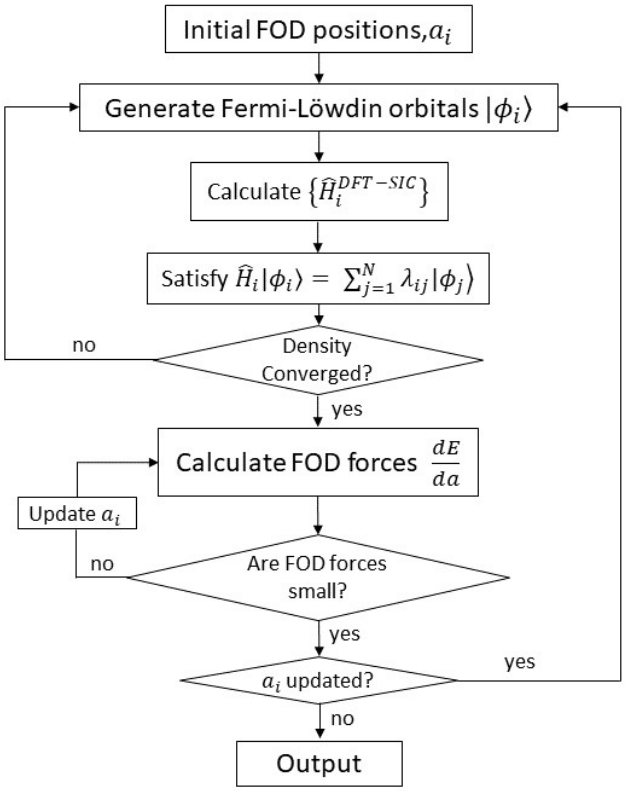


Figure 1.1: General outline of the FLO-SIC method

CHAPTER 2

METHODOLOGY

In this thesis we analyze the possibility of making the FOD optimization process faster. To make the FOD optimization process more efficient, we we focus on the current FOD optimization algorithm and try to detect which part of this algorithm is the most time consuming, and if it can potentially be made more efficient. In the algorithm depicted in Fig.1.1, the upper half is the SCF loop where the density is optimized until self-consistency is reached. The second inner loop is the part where FODs are optimized to minimize the energy at a fixed density. In this part of the algorithm, the FOD forces are calculated at each iteration and the FOD positions are then updated using these forces. At each iteration, FOD positions are updated until the forces have reached a specified minimum threshold. At each iteration, the FOD forces are evaluated using the equation

$$\frac{dE^{SIC}}{da_i} = \sum_{kl} \langle \psi_l | V_k^{SIC} | \psi_k \rangle \left[\left\langle \frac{d\psi_k}{da_i} \middle| \psi_l \right\rangle - \left\langle \frac{d\psi_l}{da_i} \middle| \psi_k \right\rangle \right] . \tag{2.1}$$

In this equation, V_k^{SIC} represents the SIC potentials which include both self-exchange correlation and self Hartee contributions. The term $\langle \psi_l | V_k^{SIC} | \psi_k \rangle$ is calculated using numerical quadrature on a grid. If we have a complex system with large number of FODs, the matrix $\langle \psi_l | V_k^{SIC} | \psi_k \rangle$ becomes very large, and calculating this matrix on each step of the iteration consumes a significant amount of time. Moreover, the k runs over the number of occupied orbitals, so that the number of potentials to evaluate grows linearly with the number of electrons in the system. If we can simplify the evaluation of the SIC potentials at each iteration step, it would make the optimization process a lot less time consuming. We propose to explore a scheme where we calculate the SIC potentials only once at the end of the SCF loop of the algorithm and then at each step of the FOD optimization process, we reuse the potentials calculated at the end of SCF loop. Within this scheme, we do not calculate the potentials at each FOD optimization step, but we just reuse the potentials calculated beforehand. Therefore, the forces calculated at a certain iteration will not

be exact for that step, i.e., for that FOD configuration, the forces calculated will be approximate forces:

$$\left(\frac{dE^{SIC}}{da_i}\right)_{approx} = \sum_{kl} \langle\psi_l|(V_k^{SIC})_{fixed}|\psi_k\rangle \left[\left\langle\frac{d\psi_k}{da_i}\middle|\psi_l\right\rangle - \left\langle\frac{d\psi_l}{da_i}\middle|\psi_k\right\rangle\right] . \tag{2.2}$$

Here, $(V_k^{SIC})_{fixed}$ is the potential calculated at the end of SCF loop which remains fixed throughout the FOD iterations.

As the $(V_k^{SIC})_{fixed}$ remains fixed, the difference in the forces at each iteration step comes from the localized orbitals ψ_k. At each step, the change in the FODs will originate in the localized orbitals, which will change the entire right side of Equation 2.2, $\left[\left\langle\frac{d\psi_k}{da_i}\middle|\psi_l\right\rangle - \left\langle\frac{d\psi_l}{da_i}\middle|\psi_k\right\rangle\right]$ and also the bra and ket of the $\langle\psi_l|(V_k^{SIC})_{fixed}|\psi_k\rangle$ matrix element. Therefore, although the potential $(V_k^{SIC})_{fixed}$ is fixed, the forces will change at each iteration step and, in principle, dictate the FOD change in subsequent iteration steps. In other words, this approximation assumes that the changes in the forces come mostly from the changes in the orbitals $\left|\frac{d\psi_l}{da_i}\right\rangle$. The forces are thus not exact, but as the iteration proceeds, the FOD positions are optimized and FODs get closer to the optimal FOD positions. As the FODs approach the optimal position, the difference between approximate and actual forces becomes smaller since approximate and exact forces are identical for the same set of FODs. Thus, at the optimal FOD positions, the approximate forces are equal to the exact forces, which is a desired feature of the approximation.

In this thesis we analyze the possibility of using the approximate forces described above for FOD optimization. To this end, we explore the approximate force landscape to determine whether they are viable for FOD optimization. We test this scheme extensively to find out if we can in principle bypass calculating potentials at every step and use approximate forces to optimize the FODs, which will in turn make the optimization process significantly faster.

CHAPTER 3

RESULTS

To test if the approximate forces are viable for FOD optimization, first the standard FLO-SIC code needed to be adapted in such a way that it takes the real-space potentials calculated on a grid at the end of the SCF loop as input, and uses them in each iteration for calculating forces (instead of re-evaluating the potentials at every step). At first, a test molecule is selected, then a standard FLO-SIC calculation is performed. In this case, we only go through the SCF loop of the FLO-SIC calculation, which gives us the potentials which can be used later. We call this calculation to generate the potentials as the *reference* calculation. Then we can use the potentials generated in the reference calculations as inputs for our future code runs. A modified version of the FLO-SIC code, which we call the Frozen Potential (FP) code, calculates the forces for a given set of FODs. Using the FP code we can calculate the FOD force for different position of an FOD and then make a plot of the force landscape for that FOD. The results of this study are divided into three parts, optimal reference results, non-optimal reference results and time optimization results.

3.1 Optimal Reference Results

In this part of the study we start from a reference calculation where the SCF part of the FLO-SIC code uses the optimal (energy optimized) FOD positions, in such a way that all the potentials generated by this reference calculation correspond to fully-relaxed FOD positions. Using these optimal reference potentials we run the frozen potential code with the position of one of the FODs slightly displaced from its optimal position. The code gives us the force as output, which corresponds to that FOD position. Then we displace the position of that FOD by a small amount and find the force for the new position. This process is repeated for different positions of that same FOD. At the end of this process, we have a landscape of the forces for that specific FOD displacement. In this case, we expect the force to be relatively large at a position away from

the optimal position and to be small near the optimal position. To compare the results of the approximate forces with the exact forces, we evaluated the exact forces for the same FODs using a standard (unmodified) FLO-SIC code.

The first molecule we study is Hydrogen Cyanide (HCN). Figure 3.1 shows the FOD config-

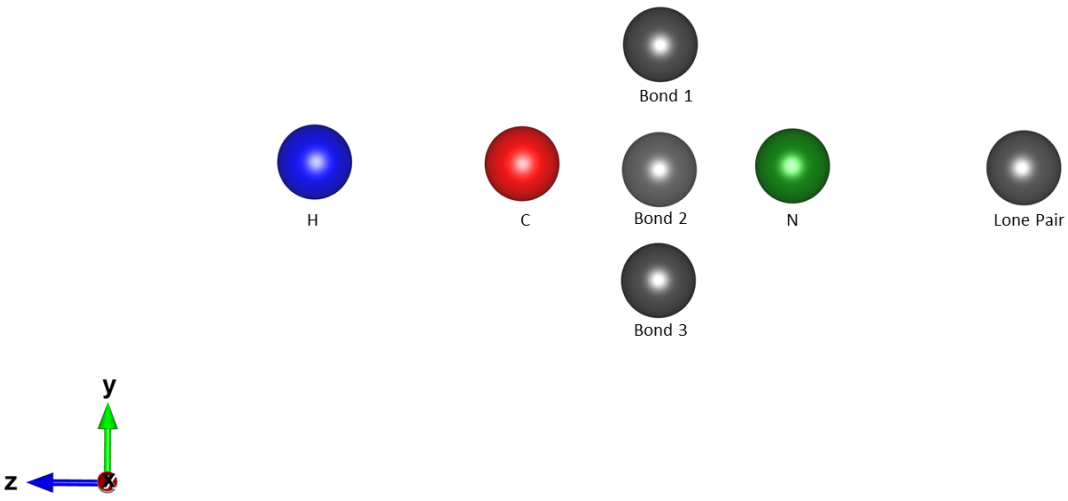
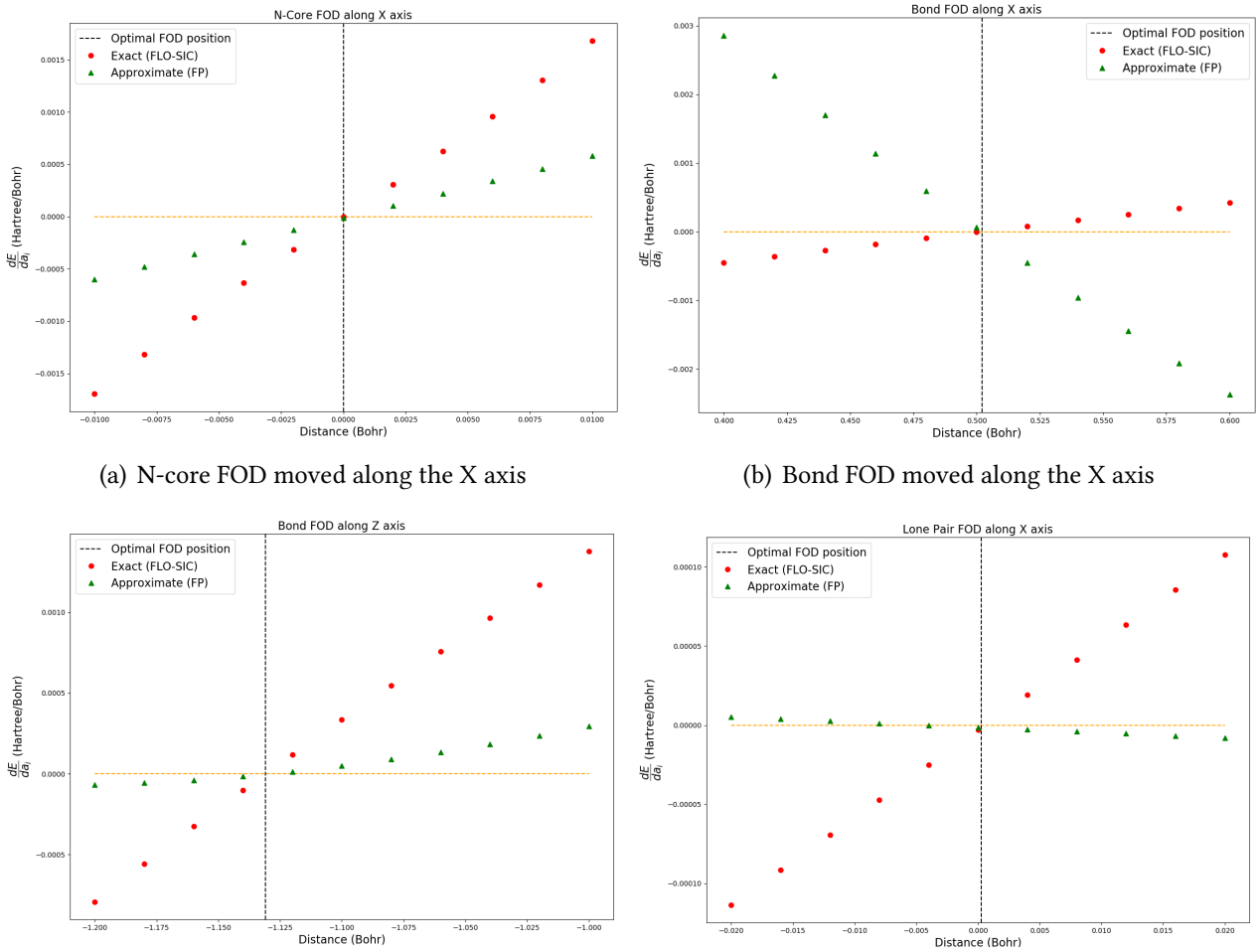


Figure 3.1: FOD configuration of the HCN molecule

uration of the HCN molecule. There are two FODs corresponding to the atomic nuclei, which we refer to as the core FODs. Then we have three FODs representing the triple bond between Carbon and Nitrogen atoms, one FOD representing the lone pair electrons on the Nitrogen atom, and one FOD representing the Hydrogen.



(a) N-core FOD moved along the X axis

(b) Bond FOD moved along the X axis

(c) Bond FOD moved along the Z axis

(d) Lone Pair FOD moved along the X axis

Figure 3.2: Force landscape for different FODs of the HCN molecule along specified directions. In the figure, the red dots represent the results from the standard FLO-SIC code, and green triangles represent the results from the Frozen Potential code. The black dashed vertical line is the optimal position of the FOD where the force is expected to be zero. The horizontal orange dashed line represent the zero forces.

The second molecule we study is the Ethylene (C_2H_4)

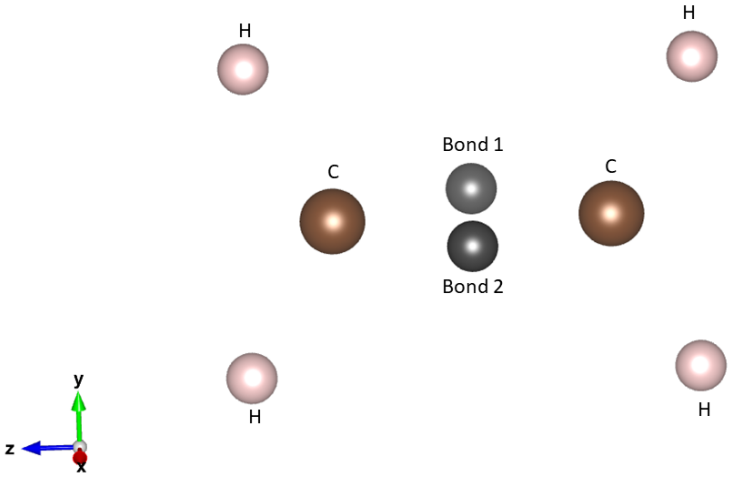
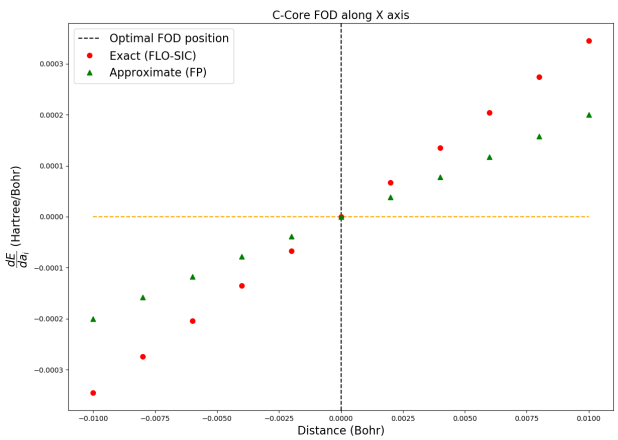
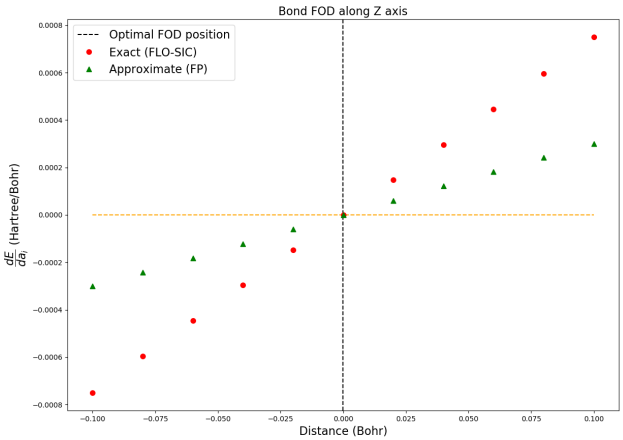


Figure 3.3: FOD configuration of the C_2H_4 molecule

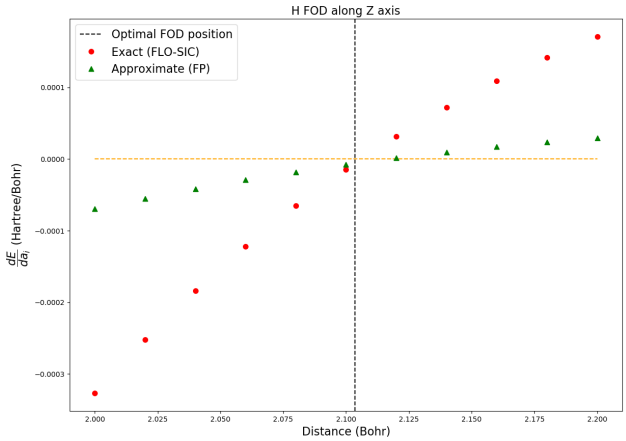
Figure 3.3 shows the FOD configuration of the C_2H_4 molecule. There are two core FODs representing the atomic nucleus of the carbon atoms, two FODs representing the double bond between Carbon atoms, and four FODs representing the C-H bond orbitals. The molecule lies on the yz plane and the two bond FODs are positioned perpendicular to the yz plane.



(a) C-core FOD moved along the X axis



(b) Bond FOD moved along the Z axis



(c) H FOD moved along the Z axis

Figure 3.4: Force landscape for different FODs of the C_2H_4 molecule along specified directions.

The third molecule we study is the Fluoromethane (CH_3F).

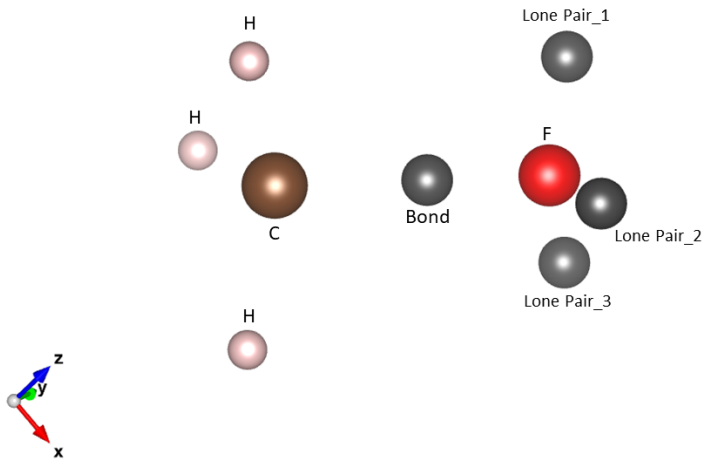
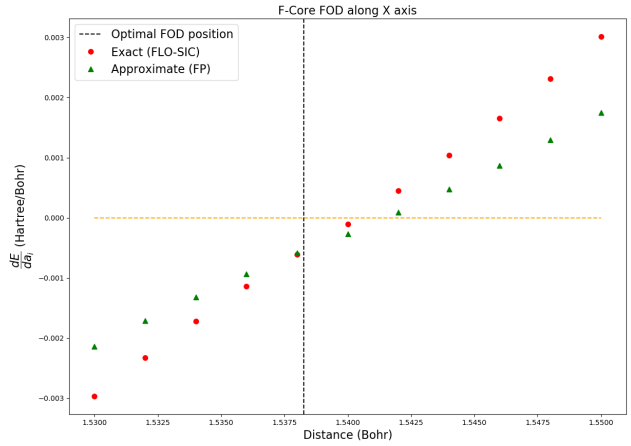
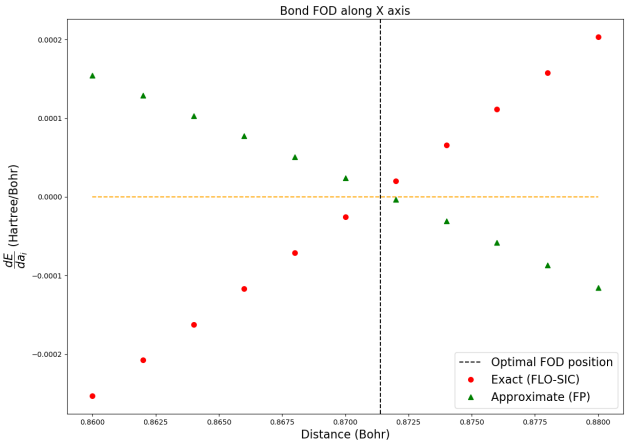


Figure 3.5: FOD configuration of the CH_3F molecule

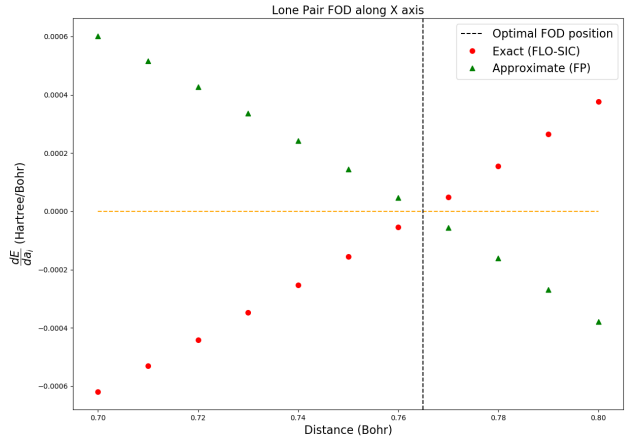
Figure 3.5 shows the FOD configuration of the CH_3F molecule. There are two core FODs representing the atomic nucleus of the carbon and the Fluorine atoms, one FOD representing the single bond between Carbon and Fluorine atoms, three FODs representing the C-H bond orbitals and three FODs representing the three lone pair electrons of the Fluorine atom.



(a) F-core FOD moved along the X axis



(b) Bond FOD moved along the Z axis



(c) Lone Pair FOD moved along the Z axis

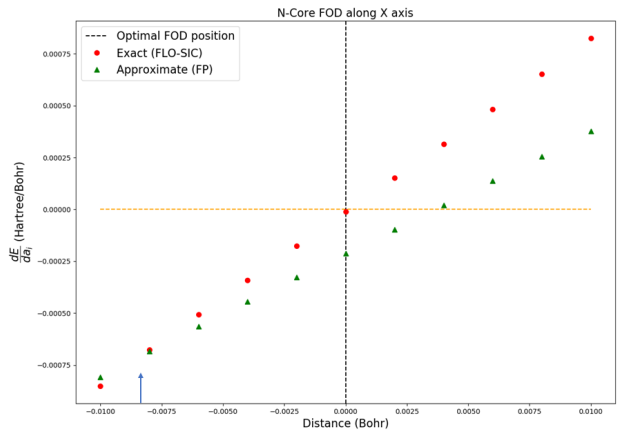
Figure 3.6: Force landscape for different FODs of the CH_3F molecule along specified directions.

From the force landscape of different FODs of these 3 molecules we can see that for the core FODs, the approximate and exact forces have the same slopes. The force is, as expected, the same for both FLO-SIC and Frozen Potential at the optimal FOD position. The difference in the forces is small near the optimal position, and becomes relatively large away from the optimal position. In the case of bond FODs and the lone pair FODs, for some cases the result is similar to the core FODs, but in some cases, for example in the cases of bond and lone pair FODs of CH_3F, for a given position the force from FLO-SIC and FP have opposite slopes. The forces still have same value at the optimal FOD position, as expected by contruction, but the opposite slopes means

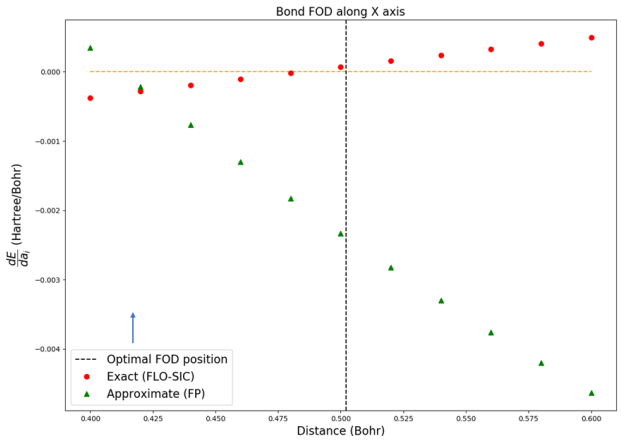
that in the FP case the FODs experience a repulsion as the approach the optimal FOD position. This is not ideal for FOD optimization process because a negative slope of the force vs distance graph means the energy is maximum at the optimal FOD position. As the optimization process searches for a minimum of the energy, in these cases where the slope is negative the optimization process will not converge. In this case, it is evident that a large contribution to the force comes from the variation of the potential originated by the difference in the localized orbitals. More investigation is required to detect the cause behind this behavior and potential fixes. In case of Hydrogen FODs, the behaviour is same as the core FODs, but the force landscape of the FP code seems is shallower than the one obtained with the FLO-SIC code.

3.2 Non-optimal Reference Results

In this part of the study we start from a reference calculation where the SCF part of the FLO-SIC code uses non-optimal FOD positions, in such a way that all the potentials generated by this reference calculation correspond to FOD positions where the FOD under investigation is slightly displaced from its optimal position. Using this non-optimal reference potentials, we repeat the same process as in the optimal reference calculations (Section 3.1) and we study the force landscape for different FODs for the three molecules studied in the previous section.

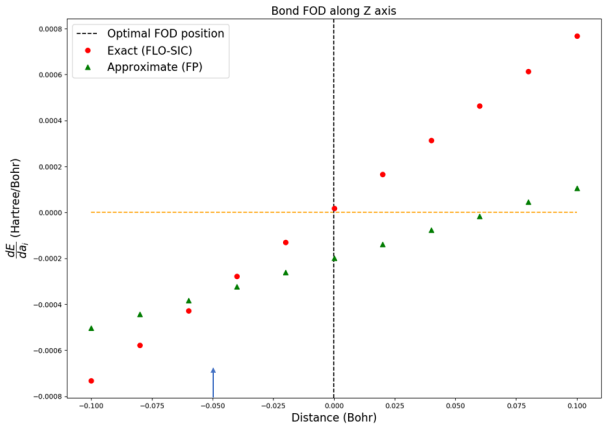


(a) N-core FOD moved along the X axis

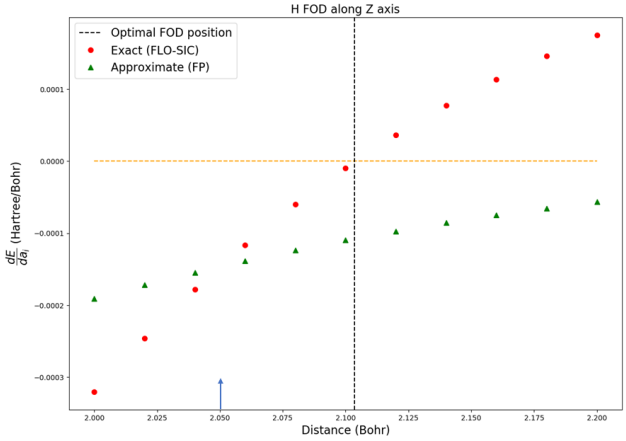


(b) Bond FOD moved along the X axis

Figure 3.7: Force landscape for two FODs of the HCN molecule along specified directions with non-optimal reference potentials. The red dots represent results from the standard FLO-SIC code, and green triangles represent the results from the Frozen Potential code. The vertical black dashed line is the optimal position of the FOD where the force is expected to be zero. The horizontal orange dashed line represent the zero force. The blue arrow indicates the position of the FOD for the reference calculation.

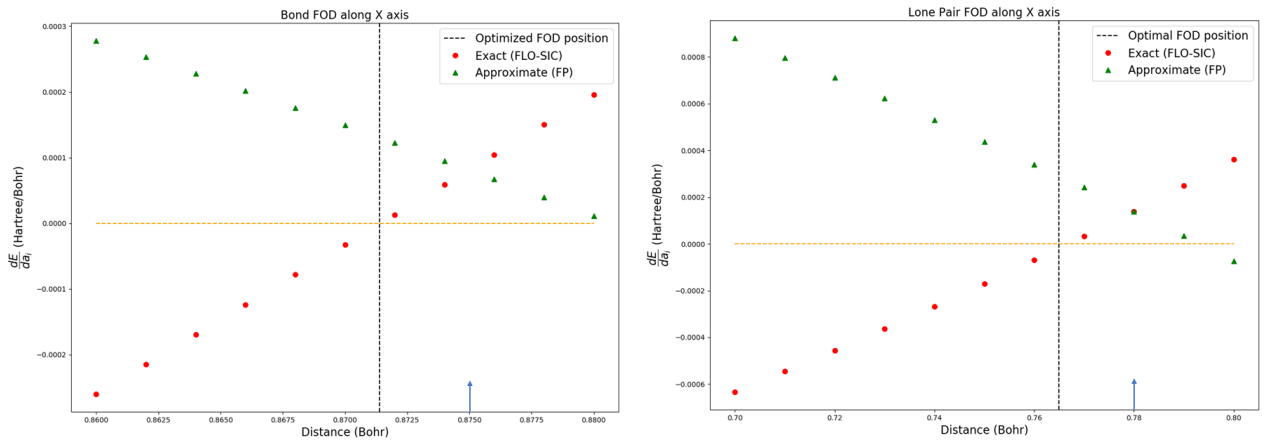


(a) Bond FOD moved along the X axis



(b) H FOD moved along the Z axis

Figure 3.8: Force landscape for 2 FODs of the C_2H_4 molecule along specified direction with non-optimal reference potentials.



(a) Bond FOD moved along the X axis

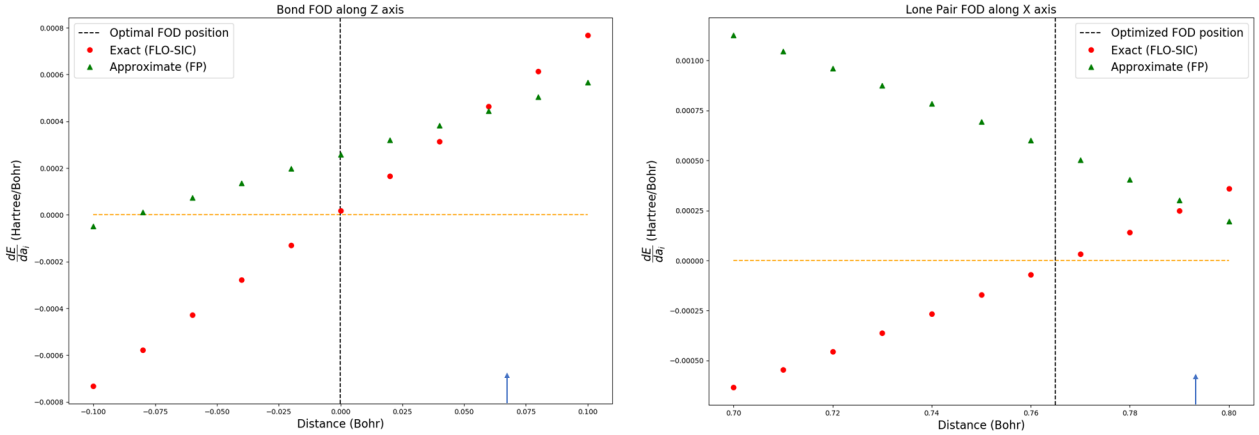
(b) Lone Pair FOD moved along the X axis

Figure 3.9: Force landscape for 2 FODs of the CH_3F molecule along specified direction with non-optimal reference potentials.

From the plots shown in Figs. 3.7, 3.8 and 3.9 we can see that the red dots or the forces from the FLO-SIC code are the same as from the optimal reference case. However, in the FP case, the forces are clearly different from the optimal reference case. The forces are not same at the optimal FOD position for FLO-SIC and FP. In the FLO-SIC case, the force is still very close to zero at the optimal FOD position, unlike in the FP case, where the force is zero at a different position then the optimal FOD. The two graphs do not coincide at the optimal FOD position, but they coincide at the position of the reference FOD, which is expected since at this position both FLO-SIC and FP have the same potentials.

Next, we changed the reference FOD positions. As we mentioned before, from the results of Figure 3.7,3.8,3.9, we can see that the FP lines cross the zero force line at a different point than the optimal FOD position. Now we try to mimic the behavior of one step of a standard optimizer using these forces. To to this, we choose the point where the FP line crosses the zero force line as the input FOD position for the reference calculation for a lone pair FOD of CH_3F and a bond FOD of C_2H_4. The results are shown in Fig. 3.8(a), where we can see that in the case of the bond FOD of C_2H_4, the FP line crosses the zero force line at 0.065 Bohr. We changed the reference FOD

position of the bond FOD to 0.065 Bohr and repeated the process. In the case of the lone pair FOD of CH_3F, from Figure 3.9(b) we can see that the FP line crosses the zero force line at 0.79 Bohr. We then changed the reference FOD position of the lone pair FOD to 0.79 Bohr and repeated the process. The idea behind this study is to see if the FP force line crosses the zero force line closer to the optimal position or not. In other words, during one step in the optimization process, if the approximate force brings the FODs closer to the exact minimum, or further away. If the later, then the approximate forces cannot be reliably used in the optimization.



(a) Bond FOD of C_2H_4 moved along the X axis

(b) Lone Pair FOD of CH_3F moved along the X axis

Figure 3.10: Force landscape bond FOD of C_2H_4 and lone pair FOD of the CH_3F molecule along specified direction with non-optimal reference potentials.

From figure 3.10 we can see that if we move the reference FOD position to a different one, then the two graphs of FLO-SIC and FP coincide at that point, which is expected by the construction of the approximate potentials. From these results it is inconclusive if the whether the approximate force brings the FODs closer to the exact minimum or further away.

3.3 Time Optimization Results

The primary objective of this thesis is to explore alternatives to make the FOD optimization process less time consuming. For this study we have calculated the time taken by both FLO-SIC

and FP method to evaluate forces for a certain FOD in 10 different positions i.e time to generate results for each graphs of the previous section. All the codes were run by the same computer using same amount of computational power.

FOD	Time taken by FLO-SIC code for 10 iterations	Time taken by FP code for 10 iterations	Time Reduction
Lone pair FOD of CH_3F along Z axis	9:54 minutes	1:45 minutes	82.5%
C-core FOD of C_2H_4 along Z axis	11.11 minutes	1:36 minutes	85.7%
Bond FOD of HCN along X axis	2:23 minutes	0:40 minutes	72.0%

Table 3.1: Comparison between the performance of FLO-SIC and FP method

From table 3.1 we can see that, with respect to the FLO-SIC code, the FP code reduces the run time by 70-80%. In case of small molecules this time reduction may seem insignificant. In case of molecules with very large number of FODs, where the FLO-SIC code may even take days to run, this reduction in run time will be very significant and may make the study with FLO-SIC a lot more efficient.

CHAPTER 4

SUMMARY

From the results presented in this thesis, we conclude that the approximate forces calculated by the Frozen Potential method follows approximately the same trend as the exact forces calculated from the FLO-SIC method, that is, the approximate forces become very close to zero at the optimal FOD positions, just like the exact force. However, one of the red flags for simply replacing the exact forces by the approximate forces is that in some cases the slope is opposite, so one would have to devise some method to overcome this. One possibility is to apply a transformation to the forces and the target function (using for example a Lorentzian) such that the zero of the transformed forces corresponds always to a minimum or a maximum of the corresponding transformed target function. Also, to employ the approximate forces on a standard optimization routine, a compatible approximate target function will be needed. Overall, this Thesis provides the first glimpse at a potential method to speed up FOD optimizations, a much needed feature for the FLO-SIC code.

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